

GenAi Training Day 2

Prompt Engineering

Overview

- Prompting is how we control LLM behavior
- Input text guides output, format, style, and reasoning
- Better prompts directly improve accuracy and reliability

What Is Prompting?

- “Talking to the model” via structured input
- LLMs follow patterns, not true understanding
- Prompts act as remote controls for the model

Prompting Roles

- **System:** sets rules, role, tone, and behavior
- **User:** instructions or questions
- **Assistant:** model-generated responses
- **Context:** previous messages and instructions

Prompting Strategies

- **Zero-shot:** no examples, relies on pretraining
- **One-shot:** single example to guide format
- **Few-shot:** multiple examples for higher accuracy

Chain-of-Thought (CoT)

- Encourages step-by-step reasoning
- Improves math, logic, and multi-step tasks
- Reduces hallucinations and errors

Tree-of-Thought (ToT)

- Explores multiple reasoning paths
- Evaluates options before choosing best solution
- Ideal for planning and complex decisions

CoT vs ToT

- **CoT:** linear reasoning, clear step-by-step logic
- **ToT:** branching reasoning, decision

Instruction Tuning & Function Calling

- Instruction-tuned models follow prompts more reliably
- Function calling enables LLMs to produce structured outputs & invoke APIs for application workflows
- Powers real-world app workflows

Prompting Do's & Don'ts

- Be clear, specific, and structured
- Use examples for complex tasks, system messages for tone, and control
- Avoid ambiguity and overloaded instructions